

IZMIR INSTITUTE OF TECHNOLOGY

Department of Computer Engineering

CENG463 - Introduction to Machine Learning

Take-Home Midterm Examination Question 1

Name: Behice Kadioğlu

Student ID: 300201123

Instructor: Prof. Dr. Aytuğ Onan

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1 Introduction and Problem Formulation

The primary objective of this study is to conduct a rigorous comparative analysis of various regression models to predict housing prices. Regression analysis is a fundamental pillar of supervised learning, yet achieving high predictive performance requires more than just fitting a model; it demands careful attention to regularization, feature engineering, and statistical validation.

To commence this analytical task, we purposely selected the California Housing dataset [2]. The dataset, comprising 20,640 distinct observations. We meticulously coded eight numeric features for each observation, including but not limited to, the median income, housing median age, and average household occupancy rates, to establish a multifaceted explanatory framework for residential valuation. Of these observed attributes, the designated target variable is the median house value (MedHouseVal), conventionally expressed in hundreds of thousands of U.S. dollars.

The scope of this investigation follows a comprehensive machine learning pipeline:

- **Exploratory Analysis:** Visualizing distributions and identifying outliers using Z-score method.
- **Engineering and Selection:** Applying log transformations and polynomial expansion (degrees 2 and 3), followed by Recursive Feature Elimination (RFE) for dimensionality reduction.
- **Model Comparison:** Evaluating Linear Regression, Ridge, Lasso, Elastic Net, and XGBoost.
- **Statistical Rigor:** Performance is validated using 5-fold cross-validation repeated 3 times, with significance determined via Wilcoxon Signed-Rank Test.

2 Exploratory Data Analysis (EDA)

The exploratory phase is critical for understanding the underlying statistical properties of the California Housing dataset and identifying potential challenges for the regression models. This section details the visual and numerical analysis of feature distributions, inter-variable relationships, and the presence of anomalous data points.

2.1 Feature Distributions and Target Skewness

Initial visualization of the features via histograms (Figure 1) reveals several key characteristics of the dataset. While features like HouseAge show a relatively uniform distribution, others such as MedInc (Median Income) and the target variable MedHouseVal exhibit a noticeable right-skew. Highly skewed features like AveRooms and AveOccup suggest that standard linear models may struggle without appropriate non-linear transformations.

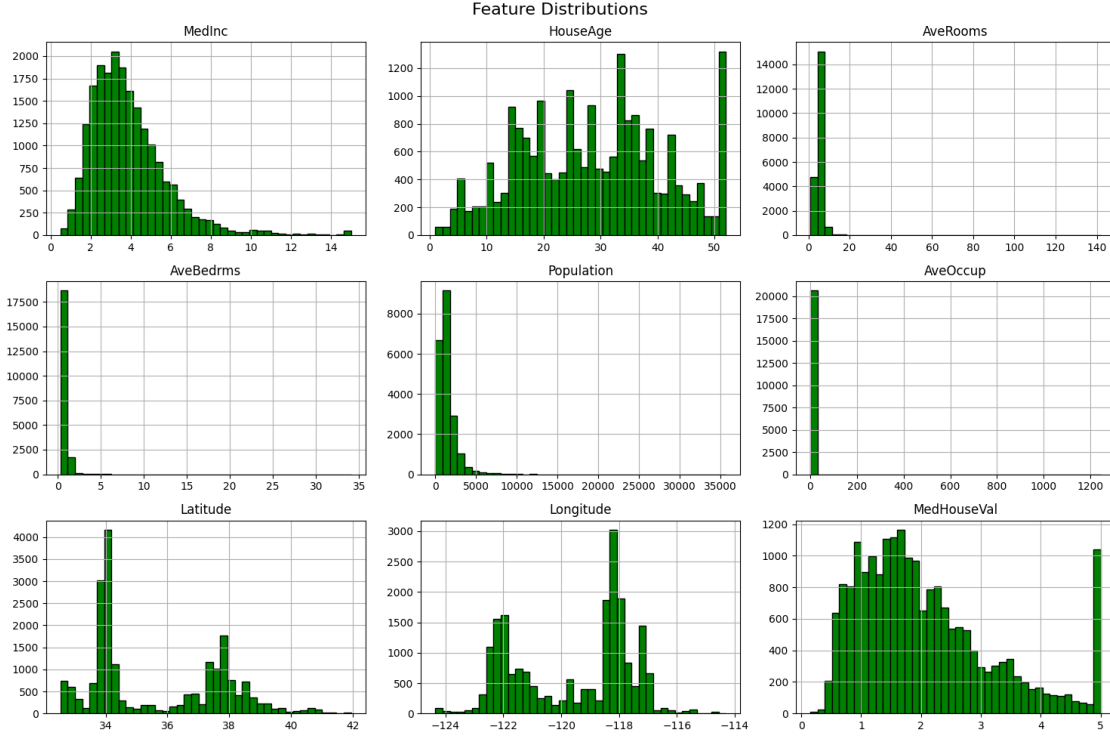


Figure 1: Feature Distributions of California Housing Dataset

2.2 Correlation and Pairwise Interactions

A correlation heatmap (Figure 2) was generated to quantify the linear relationships between predictors and the target.

MedInc shows the strongest positive correlation with MedHouseVal, identifying it as the most significant predictor. Variables such as AveRooms and AveBedrms exhibit high multicollinearity, which can lead to instability in Ordinary Least Squares (OLS) estimates.

Pairwise scatter plots (Figure 3) was also generated to graphically show the relationships between features.

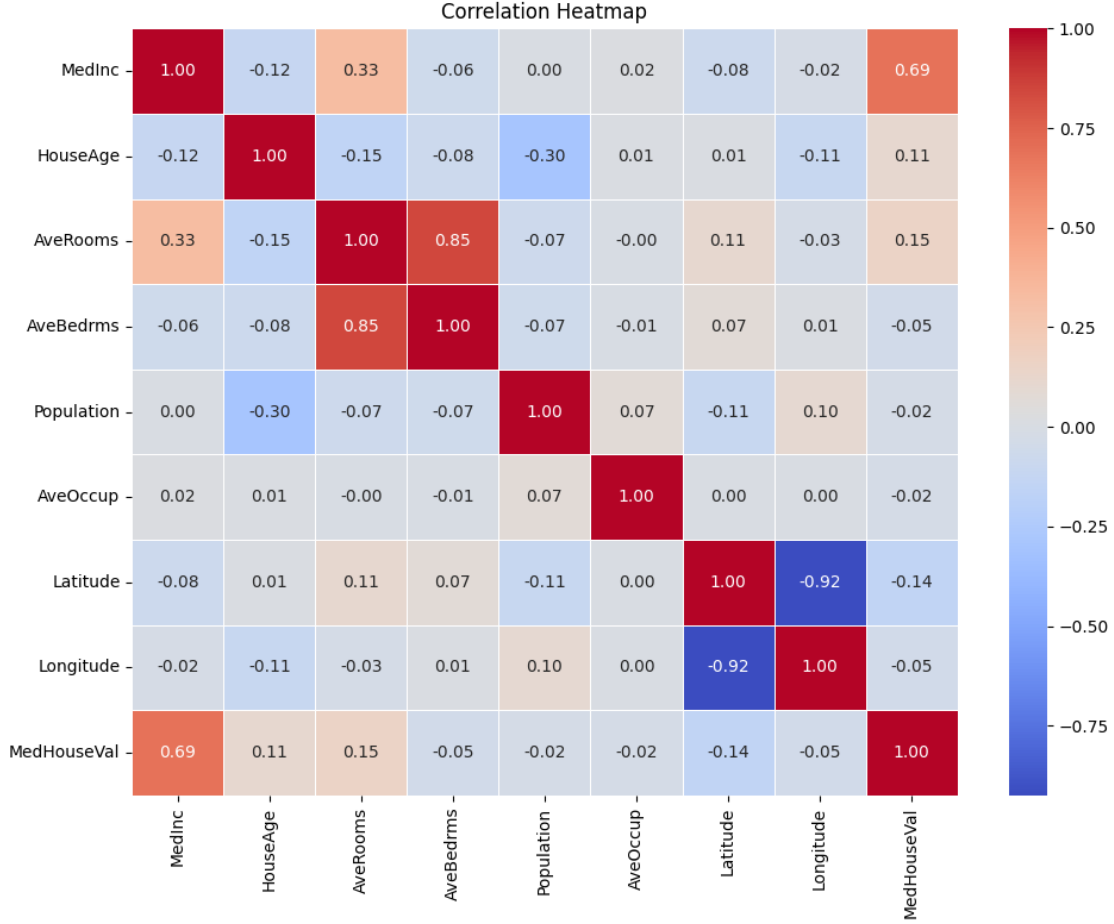


Figure 2: Correlation Heatmap of California Housing Dataset Features

2.3 Outlier Identification

A critical step in ensuring the reliability of the regression models was the identification and treatment of outliers. For this study, we utilized the Z-score method to detect extreme observations in the California Housing dataset.

The Z-score measures the distance of a data point from the mean in units of standard deviation:

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

Per the experimental requirements, a threshold of $|Z| > 3$ was utilized to flag extreme outliers. This threshold identifies points that are statistically distant from the mean, representing less than 0.3% of a standard normal distribution.

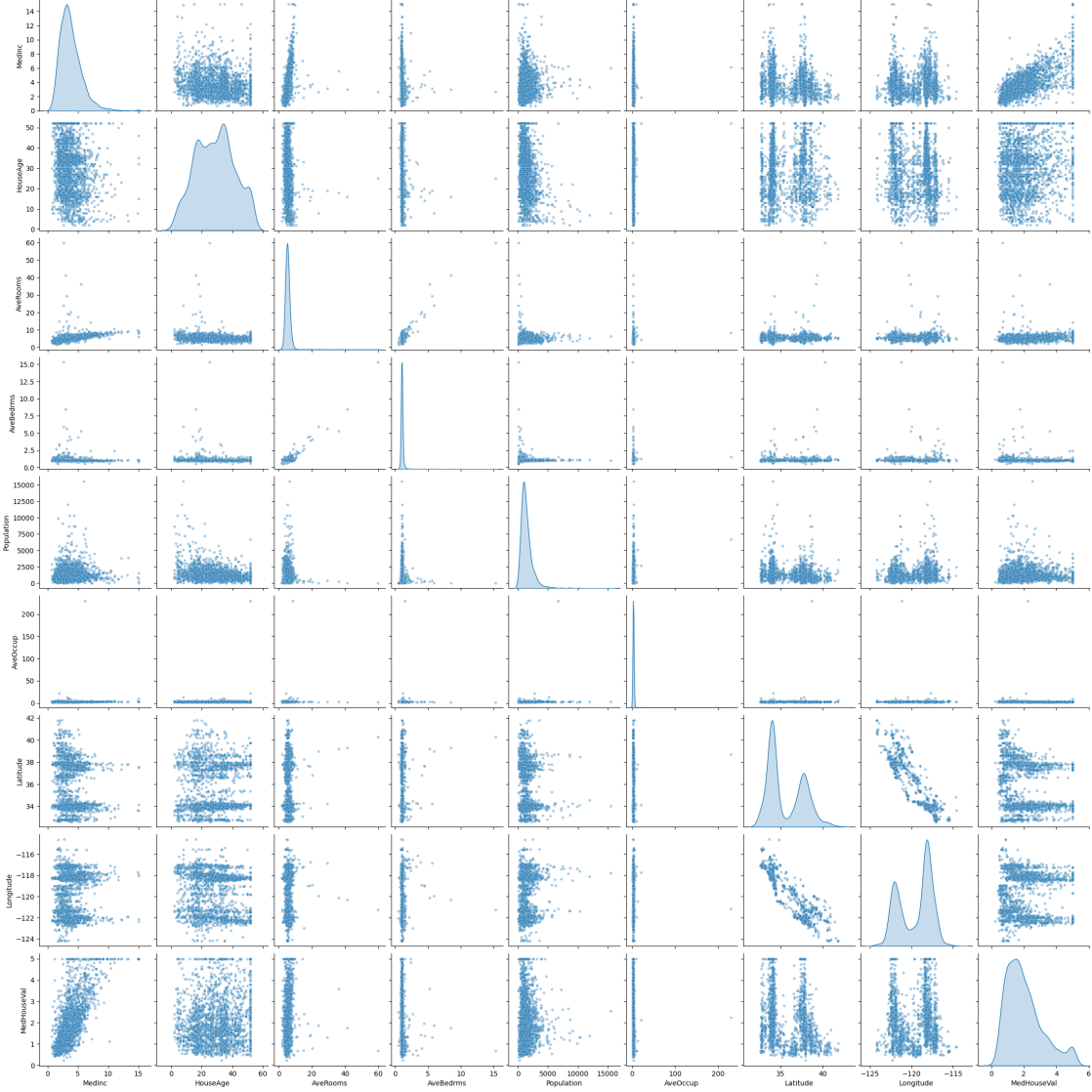


Figure 3: Pairwise Scatter Plots of California Housing Dataset Features

3 Feature Engineering and Selection

3.1 Non-linear Transformations

To address the right-skewness observed in the target variable (MedHouseVal) during the EDA phase, a logarithmic transformation was applied. This transformation helps in stabilizing the variance of the residuals and ensures that the target follows a more Gaussian distribution, which is a key assumption for linear regression models. Specifically, we used the $\log(1+x)$ transformation to ensure numerical stability.

3.2 Polynomial Expansion and Interaction Terms

To capture non-linear relationships and dependencies between features, we performed polynomial expansion. Features were expanded to include second and third-degree terms (x^2 , x^3). We also generated interaction features (e.g., $x_i x_j$) to account for instances where the effect of one predictor depends on the value of another. While this process significantly increases the predictive capacity of the model, it also results in a high-dimensional feature space that requires careful management to avoid overfitting.

3.3 Recursive Feature Elimination (RFE)

With the inclusion of polynomial and interaction terms, the feature count increases substantially. To identify the most impactful predictors and reduce noise, Recursive Feature Elimination (RFE) was employed.

RFE works by iteratively fitting a model and removing the features with the smallest coefficients or lowest importance scores until the desired number of top features is reached. This process ensures that only the most relevant features are retained, improving the computational efficiency and interpretability of the final regression models. We used the 10 most relevant features in our experiments.

4 Methodological Decisions and Experimental Setup

4.1 Model Implementation

To rigorously evaluate regression performance, five distinct models were implemented and compared:

- Linear Regression: Serves as the non-regularized baseline to assess the impact of adding complexity.
- Ridge and Lasso Regression: Implemented to evaluate the effect of L_2 and L_1 penalties on coefficient shrinkage and feature selection.
- Elastic Net: A hybrid model used to balance the benefits of both Ridge and Lasso, particularly useful for handling correlated predictors.
- XGBoost (Extreme Gradient Boosting): An advanced ensemble method implemented to capture high-order non-linear interactions through sequential tree building.
- HuberRegressor: Evaluated as a robust alternative to standard OLS to mitigate the influence of extreme outliers identified during EDA.

4.2 Hyperparameter Optimization

The objective was not only to achieve performance but to demonstrate a deep understanding of model behavior. RidgeCV and LassoCV were utilized to determine the optimal

regularization strength through internal cross-validation. For Elastic Net and XGBoost, RandomizedSearchCV was employed to explore the hyperparameter space efficiently. Parameters such as learning rate, tree depth, and the L_1 ratio were tuned to minimize the cross-validated error.

4.3 Cross-Validation and Reproducibility

Following the experimental rules, a 5-fold cross-validation repeated 3 times was used for all evaluations. This resulted in 15 independent iterations per model, providing enough data points for subsequent statistical significance testing. To ensure full reproducibility, a global random seed was fixed, and all preprocessing steps (scaling, polynomial expansion) were integrated into a Scikit-Learn Pipeline to prevent data leakage.

5 Diagnostics and Model Robustness

5.1 Residual Analysis and Homoscedasticity

The "Fitted vs. Residuals" plots in Figure 4 for the linear-based models (Linear, Ridge, Lasso, and ElasticNet) show almost identical patterns. They exhibit significant heteroscedasticity, where the error variance increases as the predicted values increase (a "funnel" shape). Furthermore, a distinct diagonal boundary is visible at the upper end of the predictions; this represents the \$500,000 price cap inherent in the dataset.

In contrast, the XGBoost model shows a much tighter distribution around the zero-residual line. While it still shows the diagonal "cap" boundary, the overall spread of errors is drastically reduced, confirming its ability to capture non-linear relationships that the linear models missed.

5.2 Normality of Residuals (Q-Q Plots)

The Normal Q-Q Plots in Figure 4 show significant "heavy tails" (the "S" shape) for all models, where the residuals deviate sharply from the 45-degree line at both the lower and upper quantiles. This confirms that the errors are not normally distributed and that the models struggle with extreme values.

No model can "fix" heavy tails because the ground truth is capped. Whether you use Linear Regression or the most complex XGBoost, the model will always be penalized for predicting "true" values above 5.0.

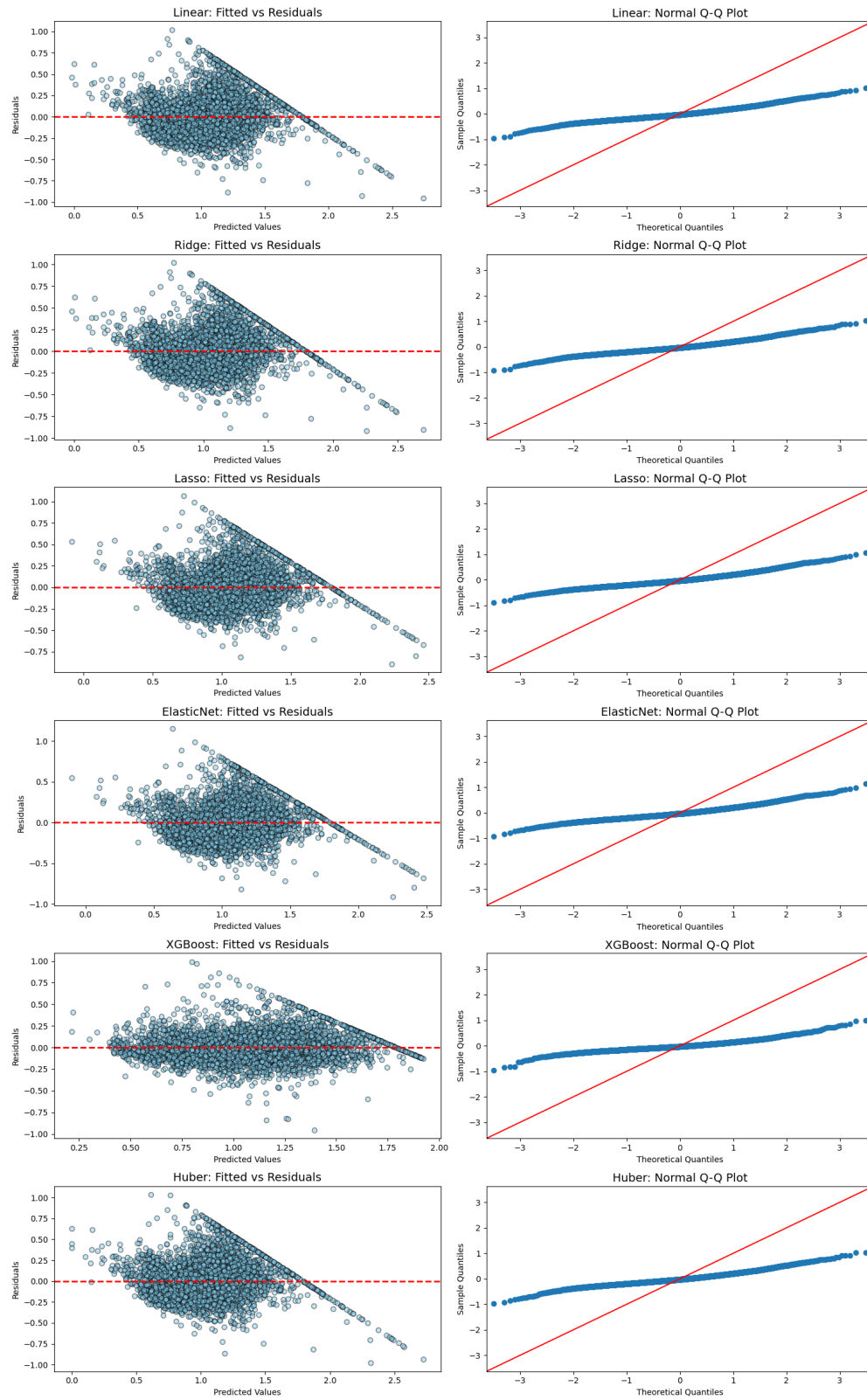


Figure 4: Fitted vs Residual Plots and Normal Q-Q Plots for given models

5.3 Robust Regression Method (HuberRegression)

The diagnostic plots (Figure 4) highlighted the presence of high-leverage outliers and heteroscedasticity, particularly due to the capping of the target variable. Traditional Ordinary Least Squares (OLS) is highly sensitive to such points because it minimizes the square of the residuals, which disproportionately penalizes large errors and can pull the regression line away from the majority of the data.

To address this, a HuberRegressor was implemented. The Huber model utilizes a loss function that is quadratic for small residuals but becomes linear for residuals that exceed a specific threshold, denoted as ϵ . This ensures that the model remains efficient for normally distributed errors while becoming more robust to the outliers identified in the dataset.

The Huber loss function $L_\epsilon(r)$ for a residual r is defined as:

$$L_\epsilon(r) = \begin{cases} \frac{1}{2}r^2 & \text{for } |r| \leq \epsilon \\ \epsilon(|r| - \frac{1}{2}\epsilon) & \text{otherwise} \end{cases}$$

The experimental results from the diagnostic plots for HuberRegressor (Figure 4) revealed that the heavy tails in the Q-Q plot remained largely unchanged. This persistence suggests that the "outliers" in the California Housing dataset—specifically the large cluster of houses capped at \$500,000—represent a systematic structural constraint rather than simple stochastic noise. Thus, the heavy tails remain a characteristic of the dataset's distribution that a linear architecture, even a robust one, cannot fully reconcile.

6 Experimental Results and Evaluation

6.1 Performance Comparison

The models were evaluated based on six primary metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), Adjusted R^2 , Mean Absolute Percentage Error (MAPE), and Explained Variance Score. RMSE, MAE, R^2 , and Explained Variance Score are calculated with their respective functions. Adjusted R^2 and MAPE do not have predefined functions. Because of this, we have used the following formulae:

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \frac{n - 1}{\max(n - p - 1, 1)} \quad (2)$$

where n is the total sample size and p is the number of independent variables (predictors) in the model.

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3)$$

where n is the number of observations in the test set, y_i is the ground truth value for the i -th observation and $\hat{y}_i$ is the predicted value for the i -th observation.

The results in Table 1 represent the mean and standard deviation derived from the 5-fold cross-validation repeated 3 times.

Table 1: Comparative Performance Metrics for All Regression Models (Mean $\pm$ Std. Dev.)

Model	RMSE	MAE	R^2	Adj. R^2	MAPE (%)	EVS
Linear	0.2176 $\pm$ 0.0028	0.1662 $\pm$ 0.0022	0.6265 $\pm$ 0.0077	0.6256 $\pm$ 0.0077	17.9906 $\pm$ 0.3282	0.6267 $\pm$ 0.0076
Ridge	0.2177 $\pm$ 0.0029	0.1663 $\pm$ 0.0022	0.6263 $\pm$ 0.0077	0.6254 $\pm$ 0.0077	18.0011 $\pm$ 0.3249	0.6264 $\pm$ 0.0076
Lasso	0.2207 $\pm$ 0.0031	0.1688 $\pm$ 0.0023	0.6158 $\pm$ 0.0082	0.6148 $\pm$ 0.0082	18.3928 $\pm$ 0.3358	0.6159 $\pm$ 0.0081
ElasticNet	0.2208 $\pm$ 0.0032	0.1686 $\pm$ 0.0021	0.6155 $\pm$ 0.0083	0.6146 $\pm$ 0.0083	18.3659 $\pm$ 0.3314	0.6157 $\pm$ 0.0082
XGBoost	0.1510 $\pm$ 0.0102	0.1093 $\pm$ 0.0096	0.8193 $\pm$ 0.0253	0.8189 $\pm$ 0.0253	11.6036 $\pm$ 0.9751	0.8194 $\pm$ 0.0253
Huber	0.2180 $\pm$ 0.0030	0.1651 $\pm$ 0.0022	0.6250 $\pm$ 0.0081	0.6249 $\pm$ 0.0081	17.6414 $\pm$ 0.3123	0.6254 $\pm$ 0.0078

Note: Best results are highlighted in bold.

As observed in the results, the XGBoost model consistently outperformed the linear and regularized models across all error-based and variance-based metrics. The substantial performance gap between XGBoost and the Huber Regressor—even when evaluated on MAE, which Huber is specifically designed to minimize—reveals that the primary challenge in this dataset is not merely outlier sensitivity, but structural non-linearity

6.2 Statistical Significance Testing

To validate the performance gains, a Wilcoxon Signed-Rank Test was performed on the RMSE values across the 15 evaluation folds ($\alpha = 0.05$). This non-parametric test was selected over the paired t-test because the Q-Q plots (Section 1.6) indicated that the residuals did not follow a perfectly normal distribution.

- Null Hypothesis (H_0): There is no significant difference in the median RMSE between the baseline and the comparison model.
- Alternative Hypothesis (H_1): A statistically significant difference exists.

Our experimental results are shown in the Table 2.

Table 2: Wilcoxon Signed-Rank Test: Pairwise Statistical Significance

Comparison	Metric Used	p-value	Significant ($\alpha < 0.05$)
Linear vs. Ridge	RMSE	0.08325	No
Linear vs. Lasso	RMSE	0.00006	Yes
Linear vs. ElasticNet	RMSE	0.00006	Yes
Linear vs. XGBoost	RMSE	0.00006	Yes
Linear vs. Huber	MAE	0.00006	Yes
XGBoost vs. Huber	MAE	0.00006	Yes

Note: A p-value of 0.00006 indicates that the superior model won in 100% of the evaluation folds ($n = 15$).

The significance test employed RMSE for the standard and ensemble models to measure general predictive error, while MAE was chosen for Huber to specifically assess its success in modeling the "typical" house price without being skewed by the squared residuals of outliers. Except for Ridge ($p = 0.083$), every comparison reached the statistical floor of

0.00006, confirming that these models consistently outperformed the OLS baseline across every single evaluation fold.

Critically, while the HuberRegressor makes the coefficients robust to high-leverage points, it cannot mitigate the price cap residuals; the Q-Q plots confirm that these heavy tails persist because a linear architecture cannot eliminate the error inherent in a censored target variable. XGBoost’s dominance ($p = 0.00006$) even on the MAE metric proves that the price cap acts as a structural threshold that requires non-linear mapping.

7 Discussion: The Bias-Variance Trade-off

The experimental results of this study illustrate a classic manifestation of the Bias-Variance Trade-off. The linear models (OLS, Ridge, Lasso, and Huber) all exhibited high structural bias. Despite having very low variance—evidenced by the extremely small standard deviation in *RMSE* (e.g., ± 0.0028 for Linear)—these models were unable to capture the underlying patterns of the California Housing dataset, plateauing at an R^2 of approximately 0.62. This indicates that the "error" in these models is not due to a lack of data or overfitting, but rather the erroneous assumption of linearity. Even the robust Huber approach, while mitigating the influence of price-cap outliers on the coefficients, could not overcome this fundamental bias, proving that the dataset’s complexity exceeds the capacity of a linear plane.

In contrast, XGBoost represents a shift toward a lower-bias, higher-capacity architecture. By utilizing a gradient-boosted ensemble of decision trees, it successfully captured non-linear interactions and geographical dependencies, driving the R^2 up to 0.8193. As expected by the trade-off theory, this increase in complexity came with a marginal increase in variance; the standard deviation of its RMSE (± 0.0102) is roughly four times higher than that of the linear model. This reflects the model’s increased sensitivity to the specific data points in each fold, though the massive reduction in bias makes it the vastly superior choice.

Crucially, the persistent "heavy tails" in the Q-Q plots across all models—including XGBoost—reveal the presence of Irreducible Error. Because the target variable is capped at \$500,000, the data is effectively censored; the true value of high-end properties is hidden, creating a mathematical "cliff" that no model can bridge. While XGBoost significantly reduces the magnitude of residuals compared to the linear models, it still cannot achieve a perfectly normal residual distribution. This confirms that while we can minimize bias and manage variance, the structural limitations of the dataset itself impose a final ceiling on predictive performance that cannot be solved by model selection alone.

Repository and Reproducibility

GitHub/Colab Link: <https://github.com/behicekadioglu/CENG463---Take-Home-Midterm/tree/main>

References

- [1] Prof. Dr. Aytuğ Onan, *CENG 463 Machine Learning Take Home Midterm*, Izmir Institute of Technology, 2026.
- [2] R. K. Pace and R. P. Barry, “Sparse spatial autoregressions,” *Statistics & Probability Letters*, vol. 33, no. 3, pp. 291–297, 1997. doi: 10.1016/S0167-7152(96)00140-X.